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An Autonomous Institute Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur

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Department of Data Science
Project Progress Seminar 1
(B. Tech 7th Semester)

On

AI Medical Report Analyzer

Guide:

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Session 2026-2027

Contents

- Introduction
- Literature Survey
- Objectives of Project
- Flow chart /Block Diagram
- Mentor's Suggestions (If any)
- Innovative Work (If any)
- Work Done (till date)
- Result & Discussion (till date)
- Project Cost Estimation
- Work Plan
- References

Introduction

Artificial Intelligence is transforming healthcare by enabling faster, more accurate interpretation of clinical data. A large majority of Electronic Health Record (EHR) data remains unstructured, existing as free-text clinical notes rather than structured, searchable fields. Medical reports are often filled with complex terminology that is difficult for patients without a medical background to understand, creating a communication gap and, in many cases, confusion, anxiety, or misinterpretation of important health information.

The AI Medical Report Analyzer converts complex medical reports into simple, patient-friendly language using OCR and NLP techniques that reliably extract clinical entities from unstructured medical text. It provides quick insights without requiring deep medical knowledge, improving accessibility and decision-making for patients.

Literature Survey			
Sr. No.	Paper Title and its Author	Details of Publication	Findings
1	Clinical entity detection in cancer reports Moreno-Casanova, J. et al.	arXiv preprint, 2025	Fine-tuned RoBERTa model on 600 reports; preprocessing improved accuracy.
2	Patient-friendly oncology summaries via LLMs Harchandani, S. et al.	medRxiv preprint, 2025	Rated 4.8/5 accuracy, 4.6/5 empathy; staging codes still confused patients.
3	Adapted LLMs outperform experts in summarization Van Veen, D. et al.	arXiv preprint, 2024	8 LLMs tested; rated equal/better than expert summaries in 81% of cases.
4	NER in electronic health records: a review Durango, M.C. et al.	Healthcare Informatics Research, 2023	Traced NER 2011-2022; BERT-based models dominant since 2016.
5	Improving extraction from pathology reports Zeng, K.G. et al.	Research Square preprint, 2023	BERT NER on 1,438 reports; F1 0.916 vs. 0.843 baseline.
6	Survey: NER & relation extraction on clinical text Bose, P. et al.	Applied Sciences, 2021	Identified data scarcity & entity ambiguity as key open challenges.

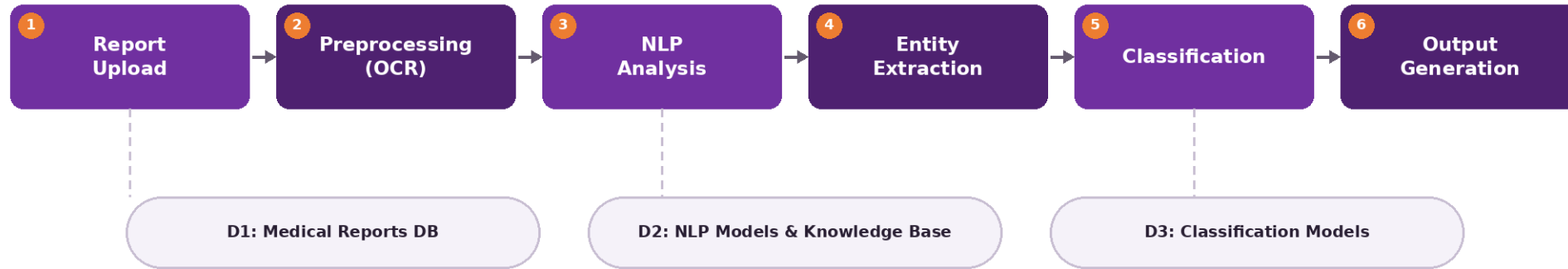
Literature Survey			
Sr. No.	Paper Title and its Author	Details of Publication	Findings
7	Speech and Language Processing (3rd ed.) Jurafsky, D. and Martin, J.H.	Stanford Univ., 2024 (Textbook)	Foundational NLP techniques underlying this project's entity-extraction module.
8	Patients' medical history summarizer using NLP Dharrao, D. et al.	Proc. IEEE Conf., 2023	T5 model produced the most coherent, information-dense summaries.

Objectives of the Project

Objective of the Project

- 1. To extract text from scanned medical reports using OCR.
- 2. To extract and classify key clinical entities -- patient details, diagnoses, medications, test results -- using NLP.
- 3. To generate a simple, easy-to-understand summary of the report.
- 4. To provide a secure platform for uploading and managing medical reports.
- 5. To develop a user-friendly web application using FastAPI and React.
- 6. To reduce manual analysis time and evaluate performance using precision, recall, and F1-score.
- 7. To maintain a history of analyzed reports, enabling patients to track health parameters over time.

Flow Chart / Block Diagram



FLOW SUMMARY

Report enters as PDF / image / text -> cleaned & standardized -> analyzed with pretrained language models -> clinical entities identified (diseases, medicines, parameters) -> report categorized by type -> patient-friendly summary generated.

Six-stage pipeline: Upload -> Preprocessing (OCR) -> NLP Analysis -> Entity Extraction -> Classification -> Output Generation.

Mentor's Suggestion (if any)

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To be updated following the guide's review at this seminar.

Innovative Work (if any)

Innovative Work

Explicit simplification of numeric and clinical codes (e.g. cancer-staging codes, lab reference ranges) within AI-generated summaries -- directly addressing a gap identified in our own literature review, where leftover technical codes were found to still confuse patients even in otherwise well-rated AI summaries.

Work done (till date)

Work done (till date)

- 1. Literature review completed; 8 papers reviewed spanning clinical NER, LLM-based summarization, and patient-facing report simplification.**
- 2. Dataset selected and explored: MTSamples corpus, 4,999 real transcribed medical reports across 40 specialties.**
- 3. Synthetic scanned-report generator built and tested: converts typed reports into realistic scanned images (with blur and skew) paired with ground-truth text, for OCR benchmarking at scale.**
- 4. Entity-extraction bootstrapping pipeline built using a pretrained biomedical NER model (scispaCy), ready to auto-label reports ahead of manual correction.**
- 5. Project repository structure and work-plan finalized across all four modules.**

Result & Discussion (till date)

No accuracy metrics have been generated yet -- entity extraction, classification, and OCR testing are still in progress. Concrete artifacts produced so far:

- Dataset profile: 4,999 reports, 40 medical specialties, average ~3,050 characters per report; heavily imbalanced (Surgery: 1,103 reports vs. several specialties under 15).
- 50 synthetic scanned-report images generated with paired ground-truth text, ready for OCR accuracy testing.
- Entity-bootstrapping script ready to run at scale using a pretrained biomedical NER model.

Next: run OCR against ground truth, and entity extraction on a sample batch, to obtain first quantitative results before the next review.

Project Cost Estimation

Sr. No.	Items	Quantity	Amount In (Rs.)
1	Development tools & libraries (spaCy, scispaCy, FastAPI, React, scikit-learn)	1 set	0 (Open Source)
2	Annotation tool (Label Studio) for entity correction	1	0 (Open Source)
3	OCR engine (Tesseract)	1	0 (Open Source)
4	LLM API credits for summarization (dev + testing usage)	Est.	2,000 - 5,000
5	Cloud hosting / deployment (backend + database, few months)	Est.	1,500 - 3,000
6	Domain / miscellaneous / testing buffer	Est.	500 - 1,000
Total Amount (Rs.)			4,000 - 9,000 (est.)

Work – Plan

Months Activities	JUNE'26	JULY'26	AUG'26	SEPT'26	OCT'26
Literature Reviews	√	√			
Dataset Collection	√	√			
OCR Development		√	√		
NLP & NER Module			√	√	
AI Summary Module				√	
Frontend and Backend Development		√	√		
Preparation of Project Report				√	√
Thesis Submission					√

References

List of papers/books/websites etc refer for project

Journals

1. Durango, M.C., Torres-Silva, E.A. and Orozco-Duque, A. (2023). Named entity recognition in electronic health records: A methodological review. Healthcare Informatics Research, 29(4): 286-297.

Text books

1. Jurafsky, D. and Martin, J.H. (2024). Speech and Language Processing (3rd ed., draft). Stanford University: Stanford, CA.

Conference proceedings

1. Dharrao, D. et al. (2023). Patients' medical history summarizer using NLP. In Proc. IEEE Conference Publication, 2023.

Journals (contd.)

1. Bose, P., Srinivasan, S., Sleeman, W.C. IV, Palta, J., Kapoor, R. and Ghosh, P. (2021). A survey on recent named entity recognition and relationship extraction techniques on clinical texts. Applied Sciences, 11(18): 8319.

References

List of papers/books/websites etc refer for project

Online journals with a DOI (Digital Object Identifier)

1. Moreno-Casanova, J. et al. (2025). Automated detection of clinical entities in lung and breast cancer reports using NLP techniques. arXiv preprint. doi:10.48550/arXiv.2505.09794.

Online journals without a DOI

1. Van Veen, D., Van Uden, C., Blankemeier, L. et al. (2024). Adapted large language models can outperform medical experts in clinical text summarization. arXiv preprint. Retrieved from <https://arxiv.org/abs/2309.07430>.

Online journals with a DOI (contd.)

1. Harchandani, S. et al. (2025). Patient-friendly summaries of oncology consultations generated by large language models. medRxiv preprint. doi:10.1101/2025.10.13.25337951; Zeng, K.G. et al. (2023). Improving information extraction from pathology reports using NER. Research Square preprint. doi:10.21203/rs.3.rs-3035772/v1.

References

List of papers/books/websites etc refer for project

Online books

1. Not applicable to this project's reference list.

Chapters from a book

1. Not applicable to this project's reference list.

Books in print form

1. Not applicable to this project's reference list.

Dissertations and Thesis

1. Not applicable to this project's reference list.

***Thank
you !***